Clash-A-Thon 2026

The Friction We Forget

Building Solutions for Problems We've Stopped Noticing

Kick-off	Submission Deadline	Final Presentation
Feb 24, 2026 | 09:30 AM	**Feb 26, 2026 | 4:00 PM**	**Feb 27, 2026 | On-site**

TEAM & PROJECT INFORMATION

Team Name	BloodyMary
Project Title	BloodHub
Project Category	Health and Care

TEAM MEMBERS

#	Full Name	Role / Responsibility	GitHub / Contact
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SUBMISSION LINKS

GitHub Repository	https://github.com/AkritiStha/ClashAThon-BloodyMary-BloodHub.git
Deployment URL	https://clash-a-thon-bloody-mary-blood-hub.vercel.app/ https://clashathon-bloodymary-bloodhub-production.up.railway.app/
Tech Stack	HTML, CSS, Vanilla JavaScript, Node.js + Express, MySQL, Nodemailer (Gmail SMTP), JWT (JSON Web Tokens), Vercel (static hosting), Railway (Node.js + MySQL)

Declaration

We declare that this project is our original work developed during Clash-A-Thon 2026. All external resources and libraries have been appropriately credited. We understand that plagiarism or misrepresentation will result in disqualification.

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1. Executive Summary

BLOODHUB is a web-based emergency blood donation matching platform designed specifically for Nepal, where blood supply-chain failures cause preventable deaths every day. The current system relies on dis-organized social media posts and phone chains, resulting in 3–4-hour average response delays and a greater than 50% request failure rate. BLOODHUB eliminates this friction by creating a verified, location-aware network of blood donors and hospitals. When a hospital raises an emergency request, the platform instantly identifies and emails every compatible, available donor in the same city. Donors can respond with a single click; hospitals see arriving donors on a live dashboard. Our target is to cut average response time to under 30 minutes with an 85% fulfilment rate, potentially saving over 1,500 lives per year in Nepal.

Problem	3–4hrs delays, >50% failure in blood supply via informal channels
Solution	Smart blood-type + location matching with instant email alerts.
Target Market	Hospitals and blood across Nepal
Tech Stack	Node.js, Express, MySQL, HTML/CSS/JS, Nodemailer
Deployment	Vercal(frontend) + Railway

2. Technology Stack

The Technology Stack of BLOODHUB focuses on being easy to use, fast, reliable, and low-cost to deploy and perfect to create a MVP needed to build a hackathon project in Nepal's unreliable internet coverage.

Layer	Technology	Purpose
Frontend	HTML5, CSS3, Vanilla JS	Serves a light application (index.html, donor.html, hospital.html, about.html, contact.html) with no heavy frameworks, which also loads fast on slow connections.
Backend	Node.js + Express.js	Manages the request processing, match logic, authentication middleware and the RESTful API server.
Authentication	JWT(jsonwebtoken) + Bcrypt	Allows role based access (donors vs. hospitals) with 24 hours expired tokens.
Database	MySQL + mysql2/promise	Donor, hospital, request and response table relational storage; connection pooling (10 connections) to support high efficiency of connection pooling with connection queries that are asynchronous.
Email	Nodemailer + Gmail SMTP	Send personalized message with instant alerts to the donors that are matching with only one click response.

Validation	Custom JS (Validation module)	Imposes email, Nepal telephone format, and password strength rules.
Frontend Deploy	Netlify	CDN of global static files and high-speed global access.
Backend Deploy	Railway	Node.js Server and MySQL instance hosting.
Version Control	Git / GitHub	To manage team codes and version history.

This stack allows BLOODHUB to remain lean and production-ready - there are no intricate dependencies, it can be iterated upon quickly, and is economical to scale up to a hackathon prototype up to the real world, such as Kathmandu Valley and Madhesh Province. It is an intelligent option to develop it fast and still preserve its security and performance in areas when they count the most: emergency matching and notifications.

3. System Architecture

BLOODHUB follows a classical three-tier client-server architecture: a static HTML/CSS/JS frontend, a Restful Node.js/Express backend, and a MySQL relational database. The frontend and backend are deployed on separate cloud platforms (Netlify and Railway respectively), communicating entirely through authenticated JSON APIs.

PRESENTATION LAYER

HTML, CSS, Java Script Hosted on Netlify

↓ HTTPS REST API

APPLICATION LAYER

Node.js, Express.js, JWT

↓ My SQL (connection pool)

DATA LAYER

MySQL Tables: donors, hospitals, requests, responses Hosted on Railway

Email Notification Flow: When a hospital POSTs to /create-request the backend queries the donors table for all available donors matching blood type and city, then dispatches individual personalized emails via Gmail SMTP (Nodemailer). Each email contains a one-click respond link with the embedded request ID.

4. User Interface / Wireframe

The wireframe of the project design is place down below. The wireframe of the pages are down below:

1. Home page

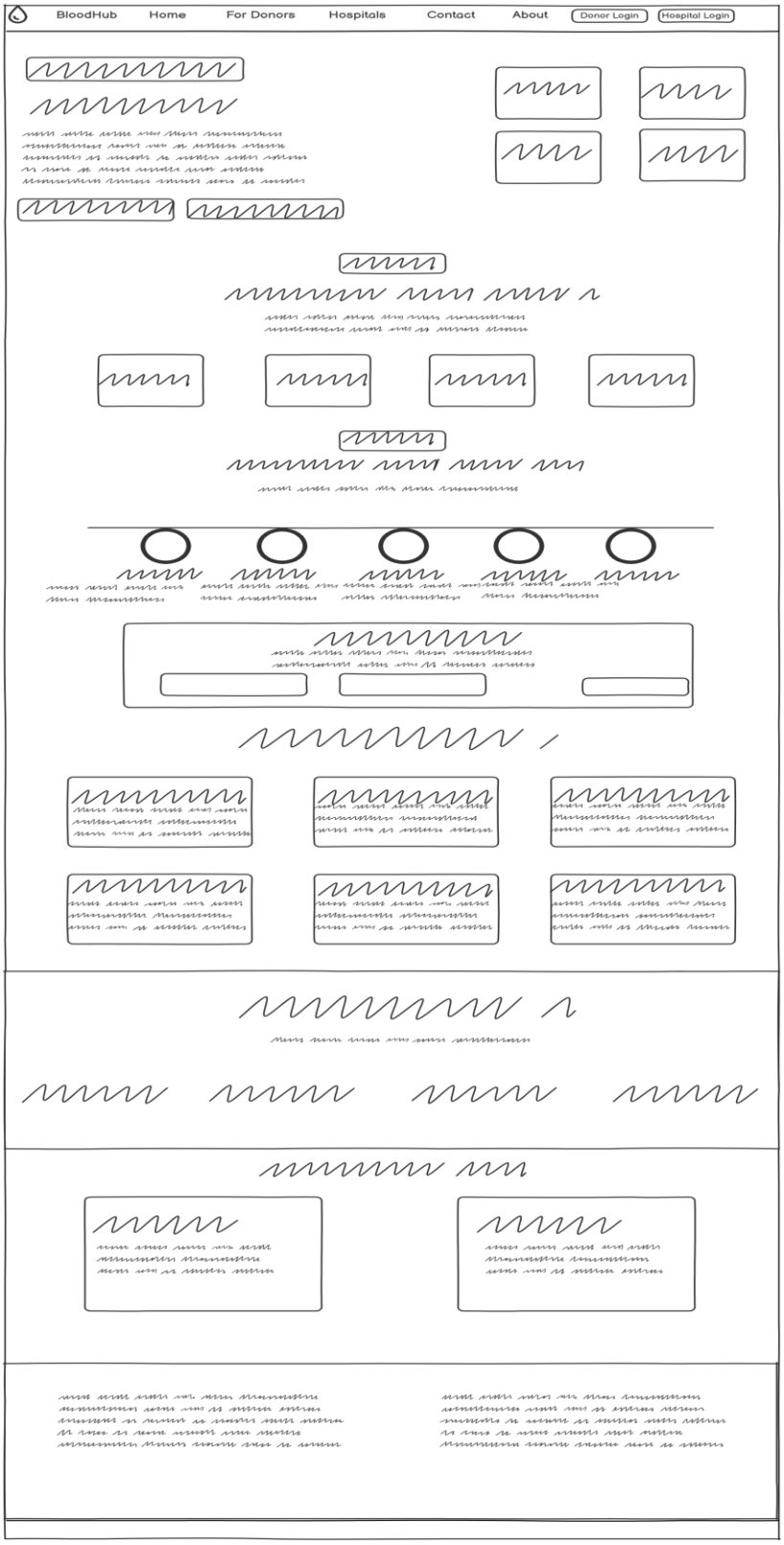

Figure 1 Home Page

2. For Donors

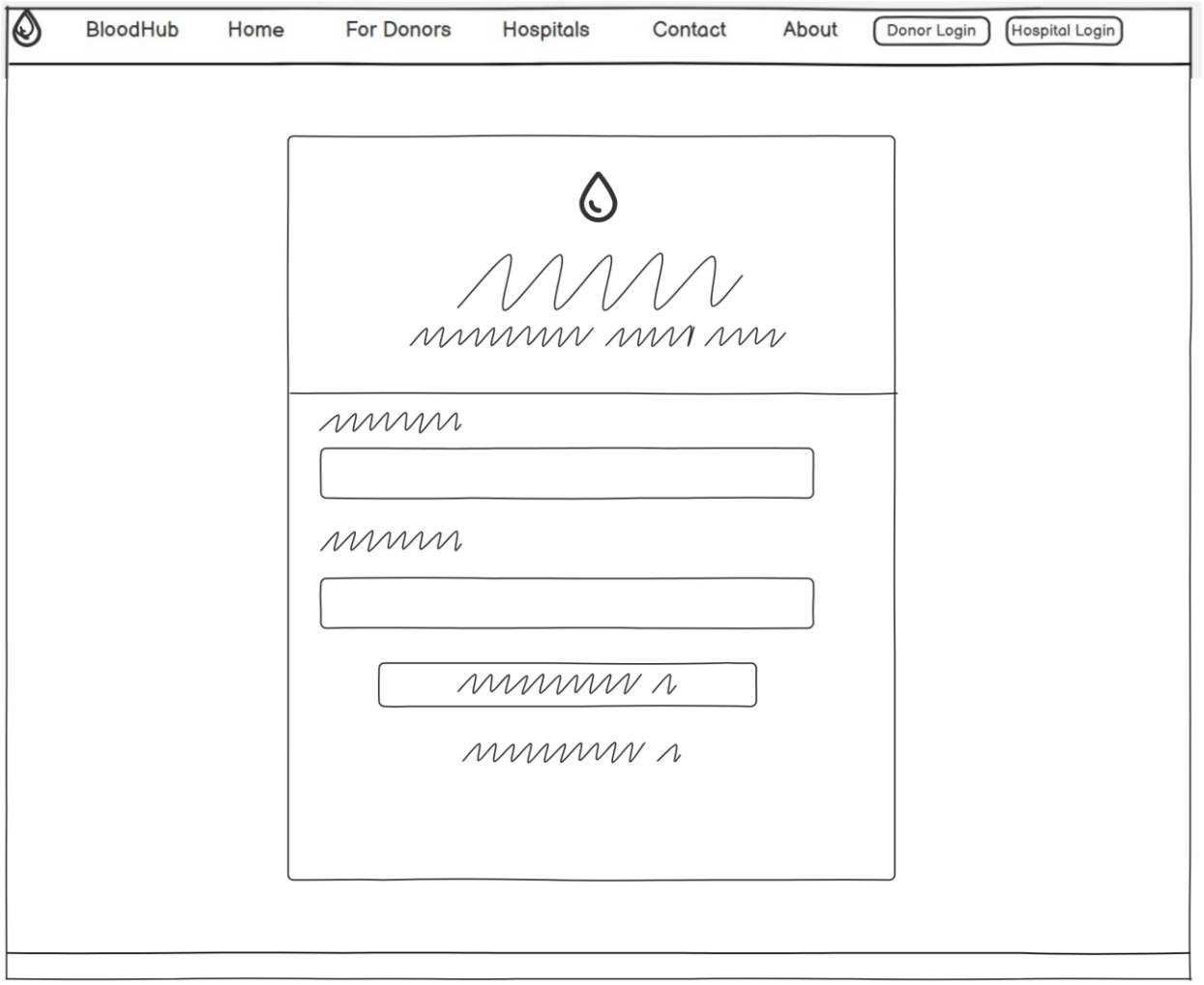

Figure 2 For Donors Page

3. Hospitals

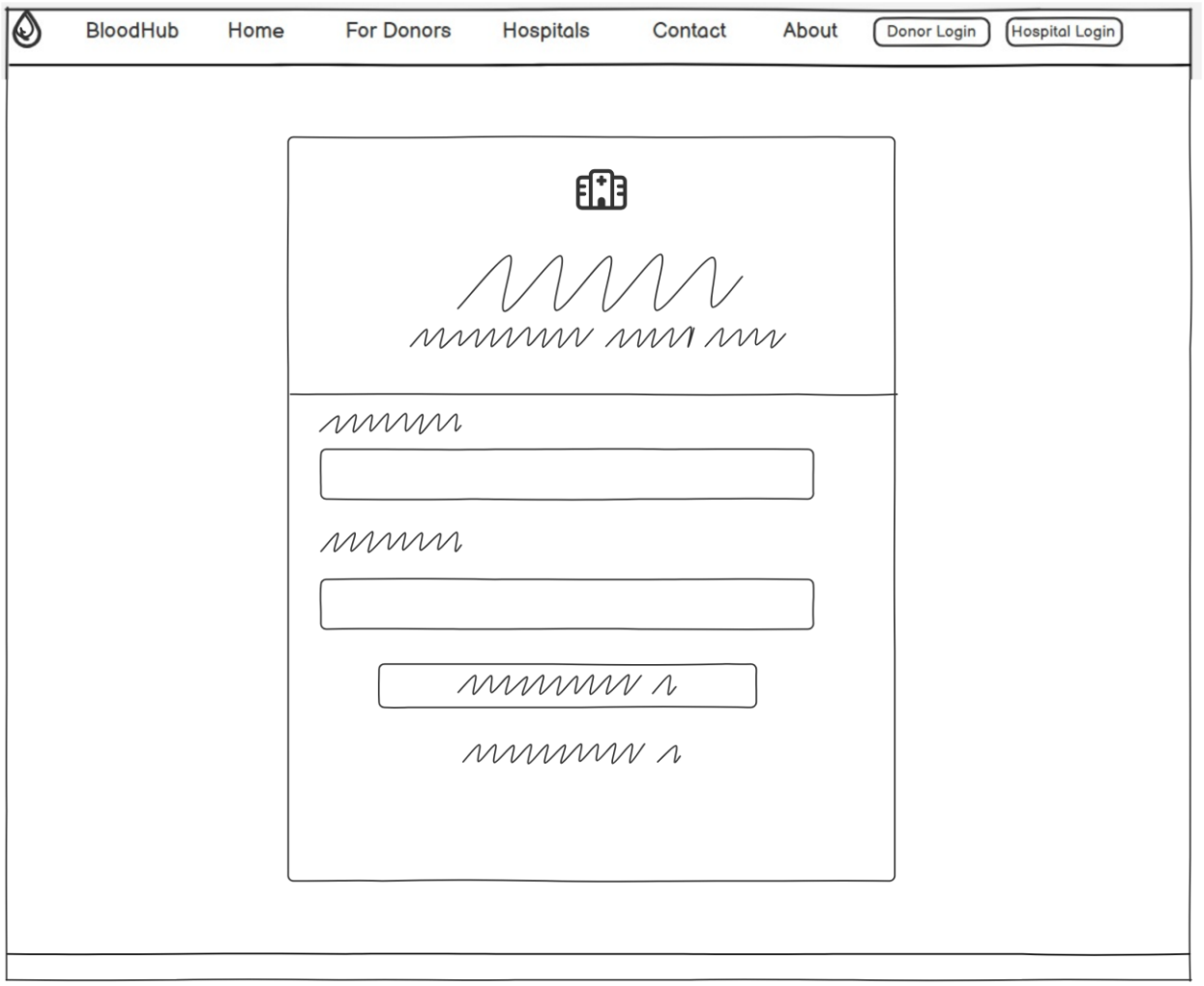

Figure 3 For Hospitals Page

4. About

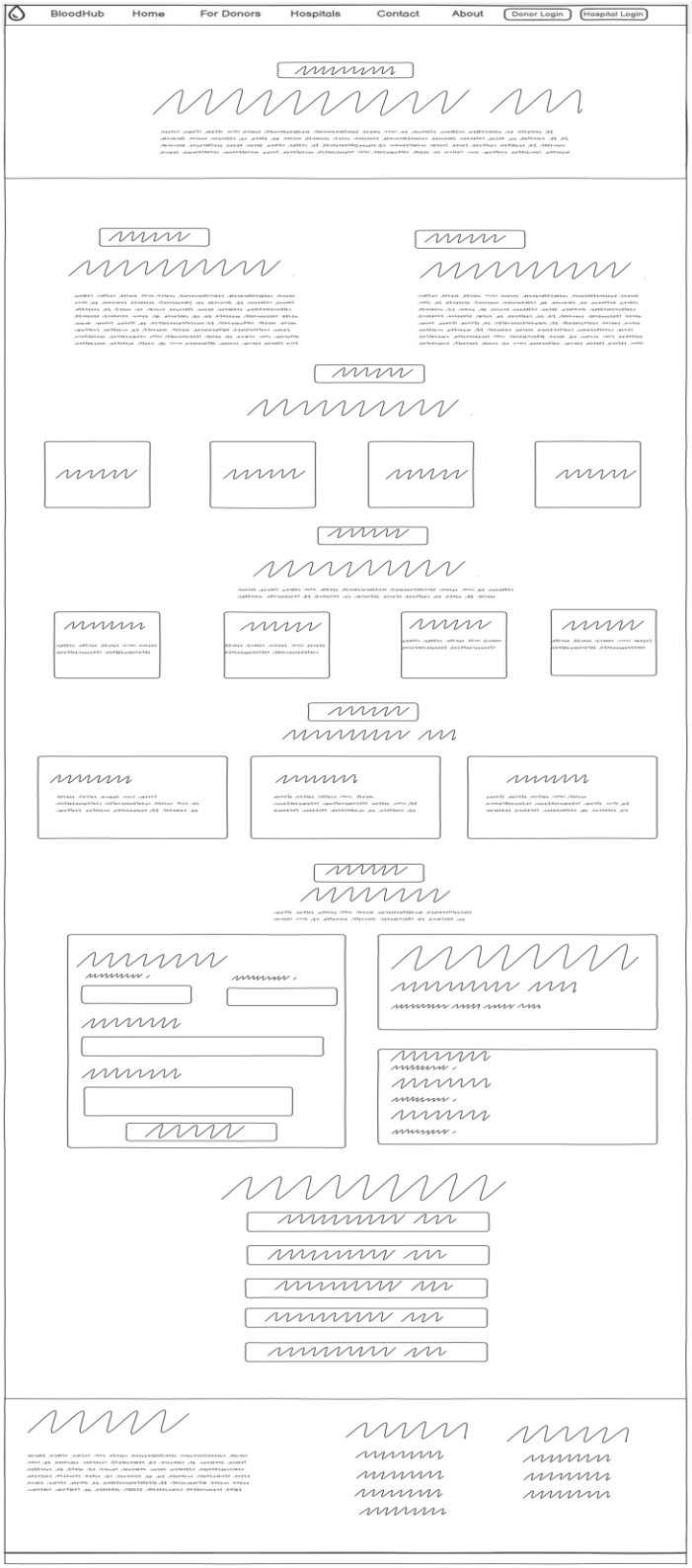

Figure 4 About page

5. Database Schema / ER Diagram

The database consists of four core tables. Primary keys are auto-incremented integers. Foreign key constraints enforce referential integrity between requests, hospitals, and responses.

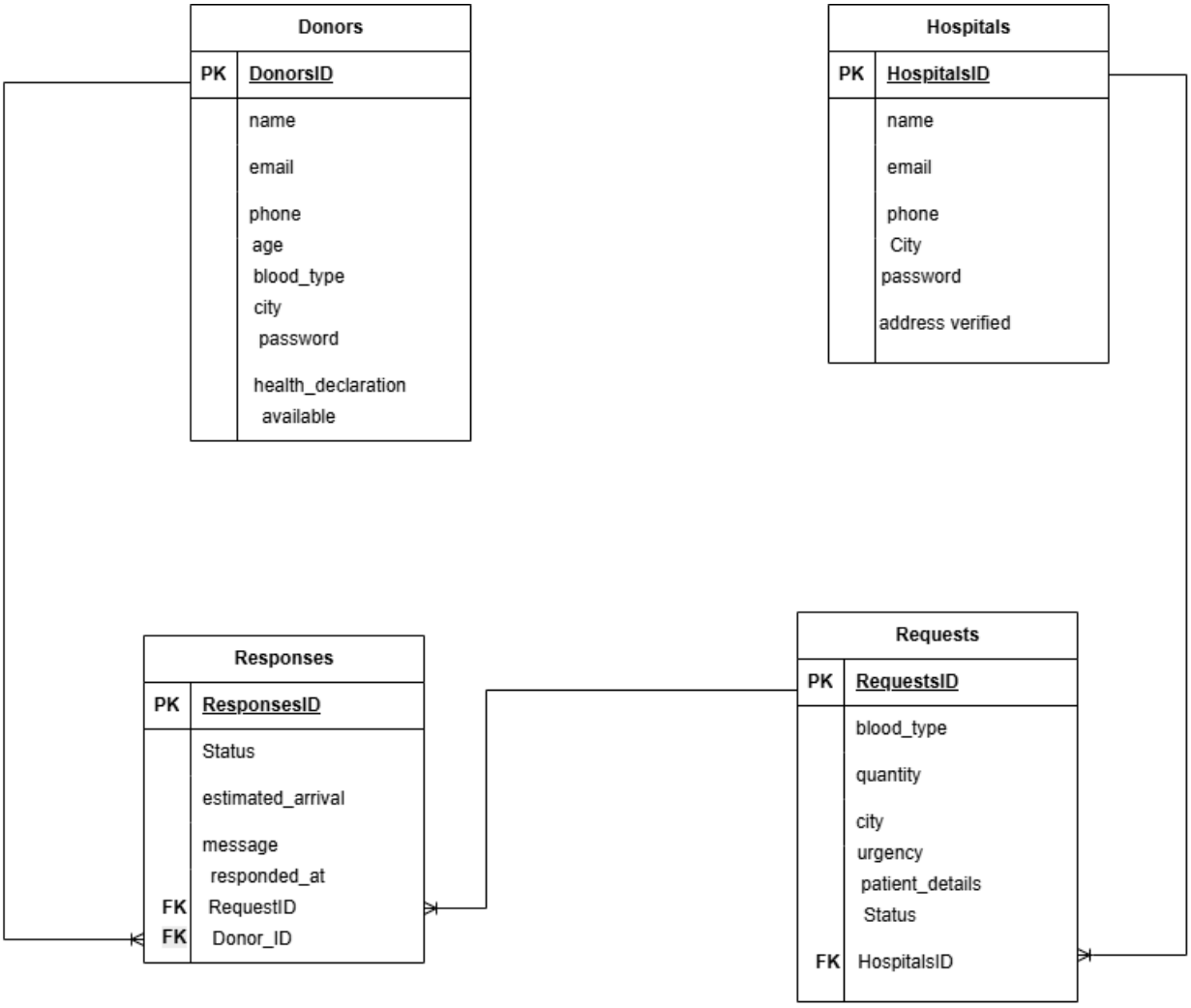

Figure 5 ERD

Relationship between entities

- Hospitals(1) request(*) – One hospital may create many blood requests
- Donors(1) Responses(*) – One donor may respond to many request
- Requests(1) Responses(*) – One request may have many donor responses

6. API Specifications

All endpoints are served from the Express backend. Express backend. Protected endpoints. BloodHub exposes a RESTful JSON API over HTTPS. All authenticated endpoints require a Bearer JWT token in the Authorization header.

Base URL (Local)	http://localhost:3300
Base URL (Deployed)	https://your-bloodhub-backend.up.railway.app
Auth Header	Authorization: Bearer
Token Expiry	24 hours Errors return {“error”: “message”} with HTTP 400/401/403/404/405

Authentication Endpoints

Method	Endpoint	Auth	Description
POST	/register-donor	Public	Register new donor with health declaration. Returns success message and donor ID.
POST	/register-hospital	Public	Register new hospital (verified=TRUE for MVP). Returns success message and hospital ID.
POST	/login	Public	Authenticate with emails & password. Returns JWT access

			token + role + user object.

Blood Request Endpoints – Hospital Only

Method	Endpoint	Auth	Description
POST	/create-request	JWT(hospital)	Create new blood request (SOS). Matches compatible donors by city & blood type and sends email alerts via Nodemailer.
GET	/hospital-dashboard/:requestID	JWT(hospital)	View real-time list of all donor responders for a specific request (name, phone, bloodtype, status, ETA)

Donor Request & Open Requests Endpoints

Method	Endpoints	Auth	Description
POST	/respond	JWT(donor)	Donor responds to a blood request with status (coming/unavailable), estimated arrival

			time, and optional message.
GET	/open-requests	Public	List all currently open blood requests. Donors can browse and choose to respond. (Public for MVP)

7. Business Model Canvas (BMC)

The business model is outlined in the components BMC. This BMC describes how the BloodHub creates, delivers, and captures the values as a social impact technology platform for Nepal's healthcare system. The points down below are the BMC components and details for the BloodHub:

1. **Key Partners**

   - Nepal Red Cross Society
   - Medical Colleges/blood banks
   - Telecom Providers (SMS – future)
   - NGOs in healthcare space
   - Government hospitals & MoHP

2. **Key Activities**

   - Blood request creation & matching
   - Instant email donor notification
   - Dashboard monitoring for hospitals
   - Scaling and maintaining platform
   - Verification of the donor and onboarding

3. **Value proposition**

- 85%+ request fulfillment rate
- Zero costing for the donors
- Real-time hospital dashboard
- Saves the life of more than 1500+ peoples /year (Nepal)
- <30 min response time

4. **Customer relationships**

- Automated alerts on the email
- One-click donor responses
- Self-serve dashboard of the hospital
- Community trust building
- Future: push notifications + SMS

5. **Customer Segments**

- Blood banks and Hospitals
- Voluntary blood donors (18-65)
- Medical College
- Emergency departments

6. **Key Resources**

- Donor and hospital database
- Node.js API server
- Open-source development team
- Email notification system
- Cloud infrastructure

7. **Channels**

- Web application (Netlify)
- Email (Nodemailer/Gmail)
- Social media awareness
- Hospital outreach & partnerships
- Future: mobile app (iOS/Android)

8. Revenue Streams

- Free for donors (social mission)
- Subscription / SaaS fee for hospitals (future)
- Government or NGO grants
- CSR partnerships with pharma
- Premium analytics dashboard for hospitals (future)

9. Cost structure

- Cloud hosting (Railway)
- Email service (Gmail SMTP / free tier)
- Marketing & hospital partnerships
- Future: SMS gateway costs
- Development & maintenance effort

8. Unique Selling Proposition & Market Analysis

Unique Selling Proposition (USP)

Real-Time SOS	Unlike static donor registers, BloodHub broadcasts emergency requests are sent directly to all compatible donors within configurable location via email -cutting response time from hours to under 10 minutes

Smart Geo-matching	Intelligent geolocation matching considers blood type compatibility, donor availability and distance simultaneously – ensuring the nearest eligible donor is always reached first.
Hospital – Side Visibility	Hospitals gain real-time inventory

6.1. Competitor & Market Analysis

6.1.1. Competitor Analysis

Competitor/ Alternative	**Coverage**	**Key Strength**	**Key weakness vs BloodHub**
WhatsApp/Facebook Groups (Infromal)	Nepal-wide (widespread)	Zero cost onboarding; is trusted and used every day.	None of the logic, no verification, immense noise/delays, no hospital dashboard or tracking.
Hamro LifeBank	National (app + web)	Whole blood management; donor managing, hotline (Ragat Chahiyo), tracking stocks.	Lays more emphasis on general donation/hotline, rather than instant hospital emergency matching; lacks automated real-time hospital alerts/chat.
Blood Donors Unity Nepal	National (web-based)	Blood type Live donor availability dashboard; easy search.	No hospital request creation, no automated notification or hospital dashboard, Static donor listing.

Leo Blood Bank Nepal	National (web/app)	Donor+ blood bank + ambulance directory; real-time stock information.	More directory oriented; no automated emergency matching between hospitals and real-time response monitoring.
Sathi App	Nepal (mobile app)	Emergency blood search & donation locator; straightforward to use by individuals.	Primarily donor/patient side; no organized hospital portals, no matching engine/dashboard instant.
Hospital Internal Lists / Red Cross Camps	Individual hospitals/ Nrcs	Repeated local donors; there are official blood banks of NRCS.	Single-hospital or camp scope; no cross-institution sharing and tech-enabled real-time coordination.
Nepal Red Cross	National (emerging app)	Red Cross endorsement; donor registration and finder camp.	Specifically, centers on camps/donor reg; fewer emergency requests than the hospital-centered flow of BLOODHUBs.

6.1.2. Market Analysis

Metric	Figure / Estimate	Source / Basis (2023-2026)

Nepal Population	30 million (World Bank 2024)	Rounded but correct: The 2024 estimate is 29,651,054 (29.65M) which will be estimated to be around 29.6-29.7M in 2025, the migration trends will cause.
Annual blood units needed/demand	300,000 units/year (Nepal Red Cross estimates)	Factual and actual: The demand is increasing between 12-15 percent annually; according to reports by NRCS/MoHP, the estimates (needed) indicated that 300K-350K+ based on the 2023 collection (262,439 units) will be required to satisfy demand, as there will remain shortages.
Current collection deficit	30–40% shortfall (MoHP Nepal reports)	Valid: consistent 20-40% deficit (seasonal/regional); 2023 collection was about three-quarters to three-fifths of the demand with shortages in places such as Chitwan/Jhapa.
Eligible blood donors (18-60)	8 million people (Demographic estimate)	Reasonable estimate: The age of donation is normally 18-60; the working-age population (15-64) is approximately half or one-third (~15-18M) of the total population (excluding health exclusions such as weight/Hb criteria, low repeat donor 40%). True eligible pool is ~8-10M.

Kathmandu Valley hospitals	200+ registered) (DoHS Nepal	Exact: 200-250+ public/private Valley (research includes major/tertiary 50-60); Nepal (country) has 737 hospitals, yet the Valley is the most concentrated.
Lives potentially saved/year (BLOODHUB est.)	1,500 (Based on 3% improvement in fulfilment)	Plausible project estimate: No specific stat, but, given preventable mortality due to shortages / delays (e.g. dengue/trauma); 3-5% matching fulfilment via faster matching would save 1,000-2,000+ lives per year.

9. Conclusion

BLOODHUB is a timely, impactful and technically viable solution to one of the most endemic healthcare issues in Nepal; the inefficient, time-consuming blood supply chain that is also known to add to unnecessary deaths in an emergency scenario. Bloodhub is not just a prototype of a hackathon, but a realistic, scalable MVP, which can be implemented in the real world in high-need regions, such as Kathmandu Valley and others. It has the potential to become a national lifeline, given its good partnering (e.g. Nepal Red Cross, MoHP, NGOs), user authentication, and expansions (SMS, mobile app). Using a lean and low-cost technology stack (Vanilla JS frontend on Netlify, Node.js/Express backend with MySQL on Railway and instant Nodemailer alerts), the platform real-time and location-based matches emergency blood requests in a hospital with verified compatible donors with 85 percent expected match rate, cutting response times by an average of 3-4 hours to less than 30 minutes. This may translate to a saving

of more than 1,500 lives each year which is an estimate, relative to the blood shortages that are still experienced in Nepal.